

In-Class Problem: Bayesian Decisions for a Drone Fleet

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Scenario

You run maintenance for a medical-supply delivery drone fleet. Before each flight, you must decide what to do about the drone's battery pack. A degraded pack can fail mid-flight and crash the drone.

Battery age. Packs are either *Old* (over 300 charge cycles) or *New*.

Quantity	Old pack	New pack
Share of fleet	30%	70%
P(degraded age)	0.20	0.02

Self-test. Every pack runs a quick onboard diagnostic that reports *Pass* or *Fail*. It behaves the same for both ages:

- $P(\text{Fail} \mid \text{degraded}) = 0.90$ (sensitivity)
- $P(\text{Fail} \mid \text{healthy}) = 0.10$ (false-alarm rate)

Actions and costs. After seeing the pack's age and the self-test result, you choose one action.

Action	Cost if healthy	Cost if degraded
Fly as-is	\$0	\$2,000 (crash: drone, cargo, cleanup)
Replace the pack	\$150	\$150
Bench-inspect, then replace only if degraded	\$60	\$210 (\$60 + \$150)

Assume bench inspection reveals the true state perfectly. Your goal is to minimize expected cost.

Questions

1. What fraction of the whole fleet has a degraded pack?
2. For each of the four cases (Old/Fail, Old/Pass, New/Fail, New/Pass), compute the posterior $P(\text{degraded} \mid \text{age}, \text{result})$.
3. Write the expected cost of each action as a function of the posterior p . Find the two thresholds on p where the optimal action switches.
4. Give the Bayes-optimal action for each of the four cases. Which result surprises you, and why?
5. Suppose you skip the self-test and decide from age alone. What is the optimal action and its expected cost for Old packs? For New packs?
6. Running the self-test costs nothing above. If it cost \$15 per flight, for which age group would you still run it? Justify with the expected value of the test.

Instructor solution

1. Fleet prior. $P(\text{degraded}) = 0.3 \times 0.20 + 0.7 \times 0.02 = 0.074$.

2. Posteriors. By Bayes' rule:

$$P(D \mid a, r) = \frac{P(r \mid D) P(D \mid a)}{P(r \mid D) P(D \mid a) + P(r \mid \neg D) P(\neg D \mid a)}$$

Case	Joint, degraded	Joint, healthy	P(result age)	Posterior p
Old, Fail	$0.20 \times 0.90 = 0.180$	$0.80 \times 0.10 = 0.080$	0.260	0.692
Old, Pass	$0.20 \times 0.10 = 0.020$	$0.80 \times 0.90 = 0.720$	0.740	0.027
New, Fail	$0.02 \times 0.90 = 0.018$	$0.98 \times 0.10 = 0.098$	0.116	0.155
New, Pass	$0.02 \times 0.10 = 0.002$	$0.98 \times 0.90 = 0.882$	0.884	0.0023

3. Expected cost and thresholds.

- Fly: $2000p$
- Replace: 150
- Inspect: $60 + 150p$

Fly beats Inspect when $2000p < 60 + 150p$, so $p < 60/1850 \approx 0.032$. Inspect beats Replace when $60 + 150p < 150$, so $p < 0.6$. The policy is: fly below 0.032, inspect between 0.032 and 0.6, replace above 0.6.

4. Optimal actions (expected costs in dollars, best in bold):

Case	p	Fly	Replace	Inspect	Decision
Old, Fail	0.692	1,385	150	164	Replace
Old, Pass	0.027	54	150	64	Fly
New, Fail	0.155	310	150	83	Inspect
New, Pass	0.0023	4.5	150	60	Fly

The surprise is Old/Pass: you knowingly fly with a 2.7% crash risk because a \$60 inspection costs more than the \$54 expected loss. A second surprise for many students is New/Fail: a failed test on a new pack is still 85% likely a false alarm.

5. Deciding from age alone.

- Old ($p = 0.20$): Fly 400, Replace 150, Inspect 90 → **Inspect, \$90.**
- New ($p = 0.02$): Fly 40, Replace 150, Inspect 63 → **Fly, \$40.**

6. Value of the test. Expected cost with the test = $\sum$ over results of $P(\text{result} \mid \text{age}) \times \text{best expected cost}$.

Age	Without test	With test	Value of test
Old	\$90	$0.26 \times 150 + 0.74 \times 54.05 = \79.00	\$11.00
New	\$40	$0.116 \times 83.28 + 0.884 \times 4.52 = \13.66	\$26.34

At \$15 per test, run it on New packs ($\$26.34 > \15) but skip it on Old packs ($\$11 < \15) and bench-inspect them directly. The test is worth more on the low-risk group, which reverses most students' intuition.

Teaching notes

The numbers are tuned so each of the three actions is optimal in at least one case, and the Old/Pass and value-of-test results land close enough to the thresholds to reward careful arithmetic. Expect 25–35 minutes for parts 1–5; part 6 works well as a stretch or a whole-class debrief.

Discussion prompts:

- How much would the crash cost have to drop before you fly Old/Fail packs? (Below about \$217, since $150/0.692 \approx 217$.)
- What if the self-test were less reliable on Old packs, say a 20% false-alarm rate? Recompute Old/Fail.
- Is minimizing expected dollars the right criterion when the cargo is medicine? Where would a risk-averse or constrained objective change the answer?

Open questions for you:

- Should students get the cost table as given, or derive the Inspect row themselves from the \$60 and \$150 components?
- Do you want a version with a continuous feature (e.g., cycle count) to bridge toward logistic-regression classifiers?